

BioCoLoop: PUBLIC-HARNESS COMPARISON

Anonymous authors

Paper under double-blind review

Public frameworks use their task-adapted native experimental controllers and the common biological fitting interface. In Panel B, AI-Scientist-v2 and AI-Researcher access laboratory 0, whereas BioCoLoop accesses ten laboratories. All Panel-B methods use Qwen2.5-7B-Instruct, the same twelve-design library and at most six candidate proposals. Scored cells pass independent recomputation from saved predictions; failure tokens record terminal controller outcomes rather than predictive scores.

Model / method	DTI	Proteomics	Cell perturbation		
	BindingDB AUROC $\uparrow$	PTPC AP $\uparrow$	VCC Top-1 $\uparrow$	Norman Top-1 $\uparrow$	Tahoe Top-1 $\uparrow$
A. Core comparison, seeds 42–44 (mean $\pm$ SD)					
Fixed model (1 lab)	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00
Qwen direct (1 lab)	89.89 $\pm$ 1.02	20.13 $\pm$ 0.07	21.09 $\pm$ 4.38	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21
BioCoLoop (10 labs)	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22	31.54 $\pm$ 0.08	<u>22.25</u> $\pm$ 0.45	22.39 $\pm$ 2.59
B. Research-controller comparison, seed 42					
Qwen direct (1 lab)	88.82	20.21	21.73	14.15	13.43
AI-Scientist-v2 (1 lab)	F_{pipe}	20.19	<u>21.73</u>	14.15	13.43
AI-Researcher (1 lab)	86.01	20.17	<u>21.73</u>	25.56	19.40
BioCoLoop (10 labs)	94.60	37.12	31.63	<u>22.74</u>	23.88

Table 1: Main results. A: core mean $\pm$ SD over seeds 42–44. B: seed-42 task-adapted public controllers use laboratory 0 and BioCoLoop uses ten laboratories. All use Qwen2.5-7B-Instruct, 12 designs, at most six candidates and 100 rounds per candidate. Scores are $\times 100$; bold/underline mark the best/second-best in each panel. F_{pipe} denotes a terminal controller-pipeline failure.

Outcome. BioCoLoop leads four of five endpoints: DTI (94.60% AUROC), proteomics (37.12% AP), VCC (31.63% Top-1) and Tahoe (23.88% Top-1). AI-Researcher leads Norman with 25.56% Top-1, versus 22.74% for BioCoLoop. AI-Scientist-v2’s DTI tree-search terminates after three trained candidates when its native substage generator fails schema validation; the table reports F_{pipe} rather than an imputed score.